CONFAI 2026 · AI IN PRACTICE (EXTENDED TRACK) · POSTER EXHIBITION

Literature Discovery with Multi-Signal Ranking, Citation-Grounded Hybrid RAG, and 2D Novelty Geometry

An End-to-End Scientific Synthesis Workstation Bridging Discovery, Comprehension, and Prior-Art Innovation

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Track: AI in Practice · 2026

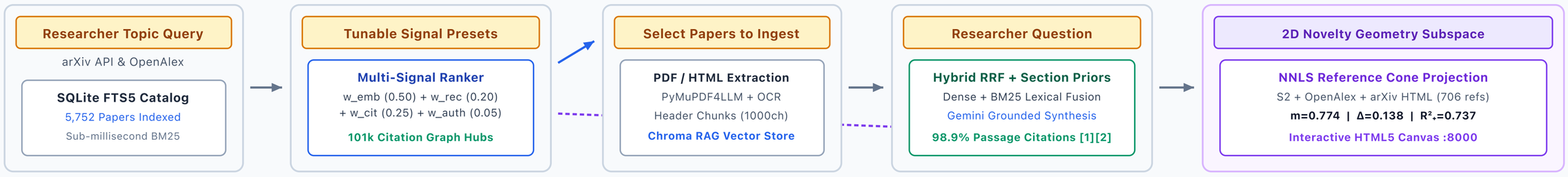
① Transparent Multi-Signal Ranking
Ranks **5,752 catalog papers** and navigates **101,716 works (226,857 edges)** using an inspectable 4-signal heuristic: semantic embedding, exponential recency decay (4.0 yr half-life), log citation saturation, and author domain continuity.

② Selective Ingest & Hybrid RRF RAG
Pure dense retrieval degrades by **41.9% in MRR** as the corpus scales to 5,663 uncurated papers. Selective full-text ingestion with dense + BM25 Reciprocal Rank Fusion and section priors preserves high precision with **98.9% citation validity**.

③ 2D Reference Cone Geometry
Non-Negative Least Squares (NNLS) cone projection into cited references decomposes paper innovation into Closest Parent (m), Multi-Parent Synergy Blend (Δ), Subspace Energy (R^2_+), and Orthogonal Novelty Residual Gap ($\sqrt{1-R^2_+}$).

END-TO-END SYSTEM ARCHITECTURE & HUMAN-IN-THE-LOOP WORKFLOW

Human Decision Point · Automated Engine · Structured Store



1. Multi-Signal Literature Discovery

RANKING & 101K GRAPH HUBS

Keyword search in unranked catalogs floods researchers with obsolete or marginally relevant preprints. Our platform computes an inspectable multi-signal utility score balancing dense semantics, time decay, citation authority, and author continuity:

$$\text{Score}(q, p) = w_{emb} \cdot S_{emb}(q, p) + w_{rec} \cdot 2^{(-\Delta t / t_{half})} + w_{cit} \cdot \frac{\log(1+C)}{\log(1+C_{max})} + w_{auth} \cdot S_{auth}(p)$$

Where t_{half} = 4.0 yr exponential decay, C = global citation count, and S_{auth} measures domain author frequency.

Weight Preset	Formula	Signal Distribution (w_emb | w_rec | w_cit | w_auth)
Balanced (Default)	0.50, 0.20, 0.25, 0.05	50% Emb · 20% Rec · 25% Cit · 5%
Precision (Focus)	0.65, 0.10, 0.20, 0.05	65% Emb · 10% · 20% Cit · 5%
Prefer Recent	0.45, 0.35, 0.15, 0.05	45% Emb · 35% Recency · 15% · 5%
Citation Heavy	0.35, 0.10, 0.50, 0.05	35% Emb · 10% · 50% Citations · 5%

101k Citation Graph Centrality: Integrates `data/citation_library/graph.sqlite3` (101,716 papers, 226,857 edges) computing in-library landmark degree hubs (e.g. *Attention Is All You Need*, *ResNet*, *Adam*) to bias discovery toward foundational literature.

2. Selective Ingestion & Hybrid RRF RAG

HIGH-PRECISION FULL-TEXT GROUNDING

Mass uncurated indexing degrades dense retrieval precision. Our workflow prompts the researcher to rank abstracts first, selectively downloading and ingesting only high-relevance full texts into Chroma:

Section-Aware Parsing
PDFs/HTML converted via **PyMuPDF4LLM** with OCR fallback. Split into 1,000 character target chunks (180 overlap) retaining markdown headings (`## Methodology`, `### Results`).

Hybrid RRF & Section Priors
Combines dense embeddings (`google/embeddinggemma-2`) and BM25 lexical matches via Reciprocal Rank Fusion. Section-intent priors boost relevant sections based on query phrasing.

$$\text{RRF_Score}(d) = \frac{w_{dense}}{60 + \text{rank}_{dense}(d)} + \frac{w_{lex}}{60 + \text{rank}_{lex}(d)} + \text{Prior}(\text{section}, q)$$

Passage-Level Verifiable Synthesis (Gemini 2.0 / 3.8 Flash)
The generator prompt strictly forces answers to anchor statements to exact passage numbers: `[1], [2]`. ConfAI evaluation measures **98.9% citation validity** (89 of 90 cited links verifiable against retrieved text chunks).

3. 2D Reference Cone Geometry

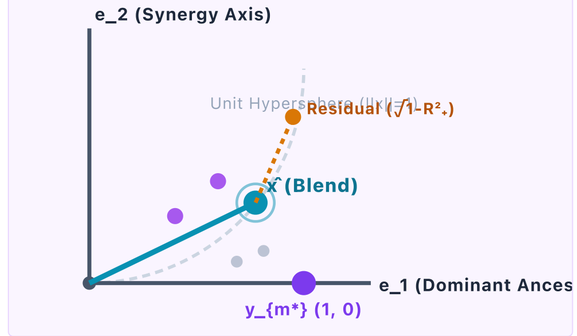
SUBSPACE INNOVATION DECOMPOSITION

Does a new paper truly break ground, synthesize disparate lineages, or simply make an incremental step? We project normalized paper vector **x** into the convex cone of its cited references **Y = [y_1, ..., y_k]** via Non-Negative Least Squares:

$$\lambda^* = \text{argmin}_{\{\lambda \geq 0\}} \| x - Y\lambda \|^2, \quad \hat{x} = Y\lambda^*, \quad R^2_+ = \frac{\| \hat{x} \|^2}{\| x \|^2} \in [0, 1]$$

Orthonormal Canvas Basis:
- **X-axis (e_1):** Aligned to closest parent y_{m^*}, plotted at (1, 0).
- **Y-axis (e_2):** Orthogonal synergy vector u / $\|u\|$.
- **Synergy Blend gain:** $\Delta = R^2_+ - \max(m, 0)^2$.
- **Novelty Residual:** $\text{Res} = \sqrt{\max(0, 1 - R^2_+)}$.

ANCHOR (M)	SYNERGY (Δ)
0.774	0.138
SUBSPACE (R^2_+)	**RESIDUAL GAP**
0.737	0.513



Real-World Discovery Validation: Evaluated on *BIG-bench* (arXiv:2206.04615). Our multi-tier citation resolver parsed **706 references** directly from arXiv HTML; NNLS attribution correctly identified *BERT* (arXiv:1810.04805) as the top foundational ancestor with **24.9% attribution share**.

4. ConfAI 2026 Empirical Evaluation

SCALING LAWS & RETRIEVAL QUALITY

Evaluating on a fixed 100-question gold benchmark reveals the **Index Scaling Law**: pure dense retrieval degrades drastically as uncurated documents are added, establishing the necessity of our selective-ingest paradigm:

Ingested Corpus Scale	Papers (N)	Chunks	Mean MRR	95% Bootstrap CI	Recall@k	Hit@1
Curated Subspace	**10**	**~1,725**	**0.786**	**[0.702, 0.863]**	**0.888**	**0.725** Peak
Topic Neighborhood	100	~9,800	0.757	[0.658, 0.842]	0.825	0.713
Domain Corpus	1,000	~98,000	0.638	[0.546, 0.740]	0.725	0.588
Full Uncurated Library	5,663	551,743	0.457	[0.360, 0.549]	0.638	0.350 -51.7%

Hybrid RRF Boost
Adding BM25 reciprocal rank fusion and section priors lifts dense MRR from **0.574 → 0.612** (+6.6%), directly recovering exact alphanumeric identifier queries.

Attribution Integrity
98.9% citation validity (89/90 cited statements verified). Rejection recall reaches **65.0%** on out-of-scope unanswerable questions.

Honest Operational Limits (AI in Practice): (1) Refusal calibration remains an open boundary: conservative safety gating leads to 23.8% false refusals on marginal contexts. (2) Alphanumeric IDs and equations underperform under pure dense retrieval without BM25 hybrid assistance. (3) 2B-parameter local embeddings require proactive Metal cache draining on Apple Silicon.

Key References: [1] Vaswani et al., *Attention Is All You Need*, NeurIPS 2017. [2] Jacot et al., *Neural Tangent Kernel*, NeurIPS 2018. [3] Cormack et al., *Reciprocal Rank Fusion*, SIGIR 2009. [4] Srivastava et al., *BIG-bench*, arXiv:2206.04615, 2022.

Inspect Source (GitHub) · Live Streamlit Demo